

Задачи по математическому анализу с SEEMOUS

1. (2007) For $x \in \mathbb{R}$, $y \geq 0$ and $n \in \mathbb{Z}$ denote by $w_n(x, y) \in [0, \pi)$ the angle (in radians) at which the segment joining the point $(n, 0)$ to the point $(n + y, 0)$ is seen from the point $(x, 1) \in \mathbb{R}^2$.

a) Show that for every $x \in \mathbb{R}$ and $y \geq 0$, the series

$$\sum_{n=-\infty}^{\infty} w_n(x, y)$$

converges. If we now set $w(x, y) = \sum_{n=-\infty}^{\infty} w_n(x, y)$, show that

$$w(x, y) \leq (\lfloor y \rfloor + 1)\pi.$$

(Here $\lfloor y \rfloor$ denotes the integer part of y .)

b) Prove that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every y with $0 < y < \delta$ and every $x \in \mathbb{R}$ we have $w(x, y) < \varepsilon$.

c) Prove that the function $w : \mathbb{R} \times [0, +\infty) \rightarrow [0, +\infty)$ defined in (a) is continuous.

2. (2008) Let $f : [1, \infty) \rightarrow (0, \infty)$ be a continuous function. Assume that for every $a > 0$, the equation $f(x) = ax$ has at least one solution in the interval $[1, \infty)$.

(a) Prove that for every $a > 0$, the equation $f(x) = ax$ has infinitely many solutions.

(b) Give an example of a strictly increasing continuous function f with these properties.

3. Let n be a positive integer and $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 x^k f(x) dx = 1$$

for every $k \in \{0, 1, \dots, n-1\}$. Prove that

$$\int_0^1 (f(x))^2 dx \geq n^2.$$

4. (2009)

a) Calculate the limit

$$\lim_{n \rightarrow \infty} \frac{(2n+1)!}{(n!)^2} \int_0^1 (x(1-x))^n x^k dx,$$

where $k \in \mathbb{N}$.

b) Calculate the limit

$$\lim_{n \rightarrow \infty} \frac{(2n+1)!}{(n!)^2} \int_0^1 (x(1-x))^n f(x) dx,$$

where $f : [0, 1] \rightarrow \mathbb{R}$ is a continuous function.

5. Let P be a real polynomial of degree five. Assume that the graph of P has three inflection points lying on a straight line. Calculate the ratios of the areas of the bounded regions between this line and the graph of the polynomial P .
6. (2010) Let $f_0 : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Define the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ by

$$f_n(x) = \int_0^x f_{n-1}(t) dt$$

for all integers $n \geq 1$.

- a) Prove that the series $\sum_{n=1}^{\infty} f_n(x)$ is convergent for every $x \in [0, 1]$.
- b) Find an explicit formula for the sum of the series $\sum_{n=1}^{\infty} f_n(x)$, $x \in [0, 1]$.
7. (2011) For a given integer $n \geq 1$, let $f : [0, 1] \rightarrow \mathbb{R}$ be a non-decreasing function. Prove that

$$\int_0^1 f(x) dx \leq (n+1) \int_0^1 x^n f(x) dx.$$

Find all non-decreasing continuous functions for which equality holds.

8. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a twice continuously differentiable increasing function. Define the sequences given by

$$L_n = \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) \quad \text{and} \quad U_n = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right), \quad n \geq 1.$$

The interval $[L_n, U_n]$ is divided into three equal segments. Prove that, for large enough n , the number

$$I = \int_0^1 f(x) dx$$

belongs to the middle one of these three segments.

9. (2012) Let $a_n > 0$, $n \geq 1$. Consider the right triangles $\triangle A_0 A_1 A_2$, $\triangle A_0 A_2 A_3$, $\dots$, $\triangle A_0 A_{n-1} A_n$, $\dots$, as in the figure. (More precisely, for every $n \geq 2$ the hypotenuse $A_0 A_n$ of $\triangle A_0 A_{n-1} A_n$ is a leg of $\triangle A_0 A_n A_{n+1}$ with right angle $\angle A_0 A_n A_{n+1}$, and the vertices A_{n-1} and A_{n+1} lie on opposite sides of the straight line $A_0 A_n$; also, $|A_{n-1} A_n| = a_n$ for every $n \geq 1$.)

Is it possible for the set of points $\{A_n \mid n \geq 0\}$ to be unbounded but the series

$$\sum_{n=2}^{\infty} m(\angle A_{n-1} A_0 A_n)$$

to be convergent? Here $m(\angle ABC)$ denotes the measure of $\angle ABC$.

Note. A subset B of the plane is bounded if there is a disk D such that $B \subseteq D$.

10. a) Compute

$$\lim_{n \rightarrow \infty} n \int_0^1 \left(\frac{1-x}{1+x} \right)^n dx.$$

b) Let $k \geq 1$ be an integer. Compute

$$\lim_{n \rightarrow \infty} n^{k+1} \int_0^1 \left(\frac{1-x}{1+x} \right)^n x^k dx.$$

11. (2013) Find all continuous functions $f : [1, 8] \rightarrow \mathbb{R}$ such that

$$\int_1^2 f^2(t^3) dt + 2 \int_1^2 f(t^3) dt = \frac{2}{3} \int_1^8 f(t) dt - \int_1^2 (t^2 - 1)^2 dt,$$

12. Find the maximum value of

$$\int_0^1 |f'(x)|^2 |f(x)| \frac{1}{\sqrt{x}} dx$$

over all continuously differentiable functions $f : [0, 1] \rightarrow \mathbb{R}$ with $f(0) = 0$ and

$$\int_0^1 |f'(x)|^2 dx \leq 1.$$

13. (2014) Consider the sequence (x_n) given by

$$x_1 = 2, \quad x_{n+1} = \frac{x_n + 1 + \sqrt{x_n^2 + 2x_n + 5}}{2}, \quad n \geq 2.$$

Prove that the sequence $y_n = \sum_{k=1}^n \frac{1}{x_k^2 - 1}$, $n \geq 1$ is convergent and find its limit.

14. (a) Prove that

$$\lim_{n \rightarrow \infty} n \int_0^n \frac{\arctan \frac{x}{n}}{x(x^2 + 1)} dx = \frac{\pi}{2}.$$

(b) Find the limit

$$\lim_{n \rightarrow \infty} n \left(n \int_0^n \frac{\arctan \frac{x}{n}}{x(x^2 + 1)} dx - \frac{\pi}{2} \right).$$

15. (2015) Prove that for every $x \in (0, 1)$ the following inequality holds:

$$\int_0^1 \sqrt{1 + (\cos y)^2} dy > \sqrt{x^2 + (\sin x)^2}.$$

16. Let $I \subset \mathbb{R}$ be an open interval containing 0, and $f : I \rightarrow \mathbb{R}$ be a function of class $C^{2016}(I)$ such that

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = f'''(0) = \dots = f^{(2015)}(0) = 0, \quad f^{(2016)}(0) < 0.$$

i) Prove that there exists $\delta > 0$ such that

$$0 < f(x) < x, \quad \forall x \in (0, \delta). \quad (1.1)$$

ii) With δ determined as in i), define the sequence (a_n) by

$$a_1 = \frac{\delta}{2}, \quad a_{n+1} = f(a_n), \quad \forall n \geq 1. \quad (1.2)$$

Study the convergence of the series

$$\sum_{n=1}^{\infty} a_n^r, \quad \text{for } r \in \mathbb{R}.$$

17. (2016) Let f be a continuous and decreasing real-valued function defined on $[0, \frac{\pi}{2}]$. Prove the inequalities

$$\int_{\frac{\pi}{2}-1}^{\frac{\pi}{2}} f(x) dx \leq \int_0^{\frac{\pi}{2}} f(x) \cos x dx \leq \int_0^1 f(x) dx.$$

When do equalities hold?

18. Let $n \geq 1$ be an integer and let

$$I_n = \int_0^{\infty} \frac{\arctan x}{(1+x^2)^n} dx.$$

Prove that

$$\text{a) } \sum_{n=1}^{\infty} \frac{I_n}{n} = \frac{\pi^2}{6}.$$

$$\text{b) } \int_0^{\infty} \arctan x \cdot \ln\left(1 + \frac{1}{x^2}\right) dx = \frac{\pi^2}{6}.$$

19. (2017) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Prove that

$$\int_0^4 f(x(x-3)^2) dx = 2 \int_1^3 f(x(x-3)^2) dx.$$

20. a) Let $n \geq 0$ be an integer. Calculate

$$\int_0^1 (1-t)^n e^t dt.$$

- b) Let $k \geq 0$ be a fixed integer and let $(x_n)_{n \geq k}$ be the sequence defined by

$$x_n = \sum_{i=k}^n \binom{i}{k} \left(e - 1 - \frac{1}{1!} - \frac{1}{2!} - \cdots - \frac{1}{i!} \right).$$

Prove that the sequence converges and find its limit.

21. (2018) Let $f : [0, 1] \rightarrow (0, 1)$ be a Riemann integrable function. Show that

$$2 \int_0^1 x f^2(x) dx \Big/ \int_0^1 (f^2(x) + 1) dx < \int_0^1 f^2(x) dx \Big/ \int_0^1 f(x) dx.$$

22. (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial function. Prove that

$$\int_0^{\infty} e^{-x} f(x) dx = f(0) + f'(0) + f''(0) + \cdots.$$

- (b) Let f be a function which has a Taylor series expansion at 0 with radius of convergence $R = \infty$. Prove that if

$$\sum_{n=0}^{\infty} f^{(n)}(0)$$

converges absolutely then

$$\int_0^\infty e^{-x} f(x) dx$$

converges and

$$\sum_{n=0}^{\infty} f^{(n)}(0) = \int_0^\infty e^{-x} f(x) dx.$$

23. (2019) A sequence $(x_n)_{n \geq 1}$ of numbers in the interval $[0, 1]$ is called a *Devin sequence* if for every continuous function $f : [0, 1] \rightarrow \mathbb{R}$ the following relation holds:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f(x_i) = \int_0^1 f(x) dx.$$

Show that a sequence $(x_n)_{n \geq 1}$ of numbers in $[0, 1]$ is a Devin sequence if and only if

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^k = \frac{1}{k+1}$$

for every $k \in \mathbb{N}$.

24. (a) Let $n \geq 1$ be an integer. Compute

$$\int_0^1 x^{n-1} \ln x dx.$$

(b) Compute

$$\sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{(n+1)^2} - \frac{1}{(n+2)^2} + \frac{1}{(n+3)^2} - \cdots \right).$$

25. (2021) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous strictly increasing function such that

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{x} = 1.$$

(a) Prove that the sequence $(x_n)_{n \geq 1}$ defined by

$$x_n = f\left(\frac{1}{1}\right) + f\left(\frac{1}{2}\right) + \cdots + f\left(\frac{1}{n}\right) - \int_1^n f\left(\frac{1}{x}\right) dx$$

is convergent.

(b) Find the limit of the sequence $(y_n)_{n \geq 1}$ defined by

$$y_n = f\left(\frac{1}{n+1}\right) + f\left(\frac{1}{n+2}\right) + \cdots + f\left(\frac{1}{2021n}\right).$$

26. For $p \in \mathbb{R}$, let $(a_n)_{n \geq 1}$ be the sequence defined by

$$a_n = \frac{1}{n^p} \int_0^n |\sin(\pi x)|^x dx.$$

Determine all possible values of p for which the series $\sum_{n=1}^{\infty} a_n$ converges.

27. (2022) Let $a, b, c \in \mathbb{R}$ be such that

$$a + b + c = a^2 + b^2 + c^2 = 1, \quad a^3 + b^3 + c^3 \neq 1.$$

We say that a function f is a *Palic function* if $f : \mathbb{R} \rightarrow \mathbb{R}$, f is continuous and satisfies

$$f(x) + f(y) + f(z) = f(ax + by + cz) + f(bx + cy + az) + f(cx + ay + bz)$$

for all $x, y, z \in \mathbb{R}$. Prove that any Palic function is infinitely many times differentiable and find all Palic functions.

28. Let $\mathcal{F}$ be the family of all nonempty finite subsets of $\mathbb{N} \cup \{0\}$. Find all real numbers a for which the series

$$\sum_{A \in \mathcal{F}} \frac{1}{\sum_{k \in A} a^k}$$

is convergent.

29. (2023) For the sequence

$$S_n = \frac{1}{\sqrt{n^2 + 1^2}} + \frac{1}{\sqrt{n^2 + 2^2}} + \dots + \frac{1}{\sqrt{n^2 + n^2}},$$

find

$$\lim_{n \rightarrow \infty} n \left(n(\ln(1 + \sqrt{2}) - S_n) - \frac{1}{2\sqrt{2}(1 + \sqrt{2})} \right).$$

30. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, strictly decreasing function such that $f([0, 1]) \subseteq [0, 1]$.

(i) For all $n \in \mathbb{N} \setminus \{0\}$, prove that there exists a unique $a_n \in (0, 1)$, solution of the equation

$$f(x) = x^n.$$

Moreover, if (a_n) is the sequence defined as above, prove that

$$\lim_{n \rightarrow \infty} a_n = 1.$$

(ii) Suppose f has a continuous derivative, with $f(1) = 0$ and $f'(1) < 0$. For any $x \in \mathbb{R}$, we define

$$F(x) = \int_x^1 f(t) dt.$$

Study the convergence of the series

$$\sum_{n=1}^{\infty} (F(a_n))^\alpha,$$

with $\alpha \in \mathbb{R}$.

31. (2024) Let $(x_n)_{n \geq 1}$ be the sequence defined by $x_1 \in (0, 1)$ and

$$x_{n+1} = x_n - \frac{x_n^2}{\sqrt{n}} \quad \text{for all } n \geq 1.$$

Find the values of $\alpha \in \mathbb{R}$ for which the series $\sum_{n=1}^{\infty} x_n^\alpha$ is convergent.

32. For every $n \geq 1$ define x_n by

$$x_n = \int_0^1 \ln(1 + x + x^2 + \dots + x^n) \cdot \ln \frac{1}{1-x} dx, \quad n \geq 1.$$

a) Show that x_n is finite for every $n \geq 1$ and $\lim_{n \rightarrow \infty} x_n = 2$.

b) Calculate $\lim_{n \rightarrow \infty} \frac{n}{\ln n} (2 - x_n)$.

33. (2025) Calculate

$$\lim_{n \rightarrow \infty} n \int_0^\infty e^{-x} \sqrt[n]{e^x - 1 - \frac{x}{1!} - \frac{x^2}{2!} - \dots - \frac{x^n}{n!}} dx.$$

34. Let (a_n) be a monotone decreasing sequence of real numbers that converges to 0. Prove that

$$\sum_{n=1}^{\infty} \frac{a_n}{n}$$

is convergent if and only if the sequence $(a_n \ln n)$ is bounded and

$$\sum_{n=1}^{\infty} (a_n - a_{n+1}) \ln n$$

is convergent.

35. (2026) Determine all differentiable functions $f : \mathbb{R} \rightarrow [0, \infty)$ with continuous derivative such that $f^2(x) \leq f'(x)$ for all $x \in \mathbb{R}$.